

Test-Time Strategies for LLMs: Conditional Pass@N and Search-Aligned Training



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Background: The Training–Inference Misalignment

Modern LLM deployment increasingly relies on **test-time strategies** (TTS) that generate multiple candidate solutions and aggregate them:

- **Pass@N**: generate N samples, succeed if *at least one* is correct.
- **Majority Vote**: select the most frequent answer among N samples.
- **Best-of- N** : use a verifier to pick the highest-scoring candidate.

However, standard training (SFT / RL) optimizes single-sample likelihood (pass@1). This creates a **misalignment**: overconfident models concentrate probability on a few answers, collapsing the output diversity that pass@ N needs [2].

Research Questions

1. Can we implement **conditional (dependent) sampling** at test time to avoid wasting budget on repeated wrong answers?
2. What **distributional conditions** make conditional search most effective?

Experiment 1: Conditional Prompting with GPT-2

We fine-tuned **GPT-2** (SFT, 2 epochs, lr = 3e-5) on **1,000 problems from the MATH benchmark** [2] and evaluated on 100 held-out problems with pass@10.

Three strategies were compared under the same sample budget:

- **Normal baseline**: 10 independent samples per problem.
- **Simple conditional**: feed back rejected answers in the prompt and ask the model to try a different approach.
- **Two-stage chunked**: generate a batch normally, then generate remaining samples conditioned on prior failures.

Strategy	pass@1	pass@10
Normal baseline	0.00	0.01
Simple conditional	0.01	0.03
Two-stage chunked	0.00	0.05

Observation

Conditional strategies showed consistent improvement over the normal baseline on pass@10 (up to 5×). However, the sample size is small and absolute accuracy is low, so these results are **suggestive but not conclusive**.

Challenges of Prompt-Based Conditioning

Prompt-based conditioning showed promise but faces practical challenges. Telling a model “do not repeat these answers” can trigger the **Pink Elephant Effect** [1]: mentioning forbidden content in the prompt paradoxically *increases* the model’s likelihood of generating it.

- Appending failure history consumed context window and added noise.
- GPT-2’s limited instruction-following made “avoid prior answers” unreliable, especially with long feedback prompts.
- The conditioning problem is **nontrivial in a Chain-of-Thought context**: even capable models struggle to reliably suppress previously seen reasoning paths.

Motivation: these practical challenges led us to study the *operator-level* properties of conditional search — under what conditions *should* it help, independent of prompt engineering?

Experiment 2: Operator-Level Validation

To isolate the search operator from prompt-following issues, we built an **exact Sampling Without Replacement (SWOR) evaluator** using a synthetic categorical distribution:

- One correct answer with probability p ; 8 rival answers sharing mass $1 - p$.
- $p \in \{0.05, 0.10, 0.20, 0.35\}$; Dirichlet $\alpha \in \{0.2, 0.5, 1, 2, 5\}$.
- Budgets $N \in \{2, 3, 4, 5\}$; 400 trials per setting; seed = 42.

The core evaluator computes **exact conditional pass@N** by recursively removing failed rivals (SWOR), then compares against independent pass@N = $1 - (1 - p)^N$.

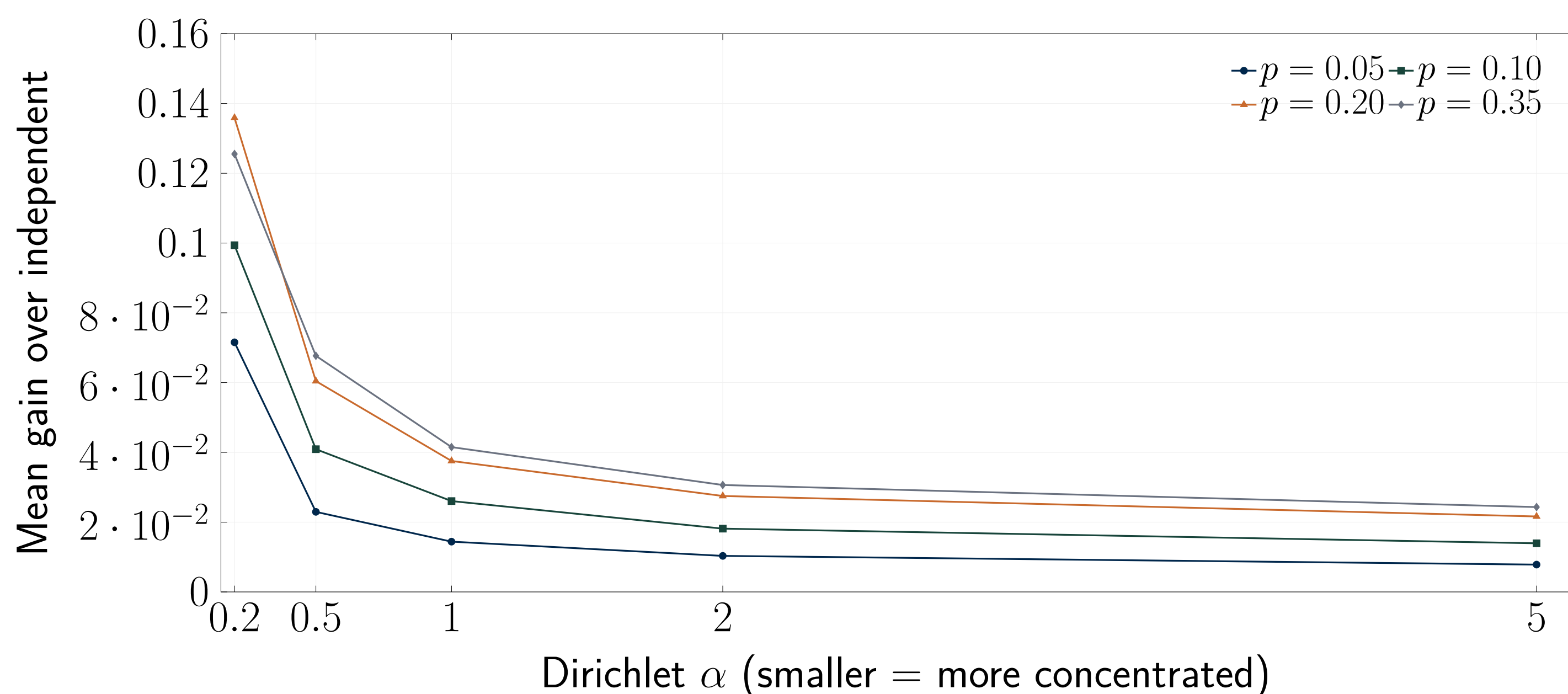
Theoretical intuition

For $N = 2$, the conditional success probability is:

$$\tilde{p}_{\text{cond}}(2) = p + \sum_{y' \in \mathcal{R}} q_{y'} \left(\frac{p}{1 - q_{y'}} \right)$$

When rival mass is **concentrated** on few answers, eliminating a likely rival yields large renormalization gains. When rivals are **diffuse**, the gain is minimal.

Result 1: Concentrated Error Mass Amplifies Conditional Search

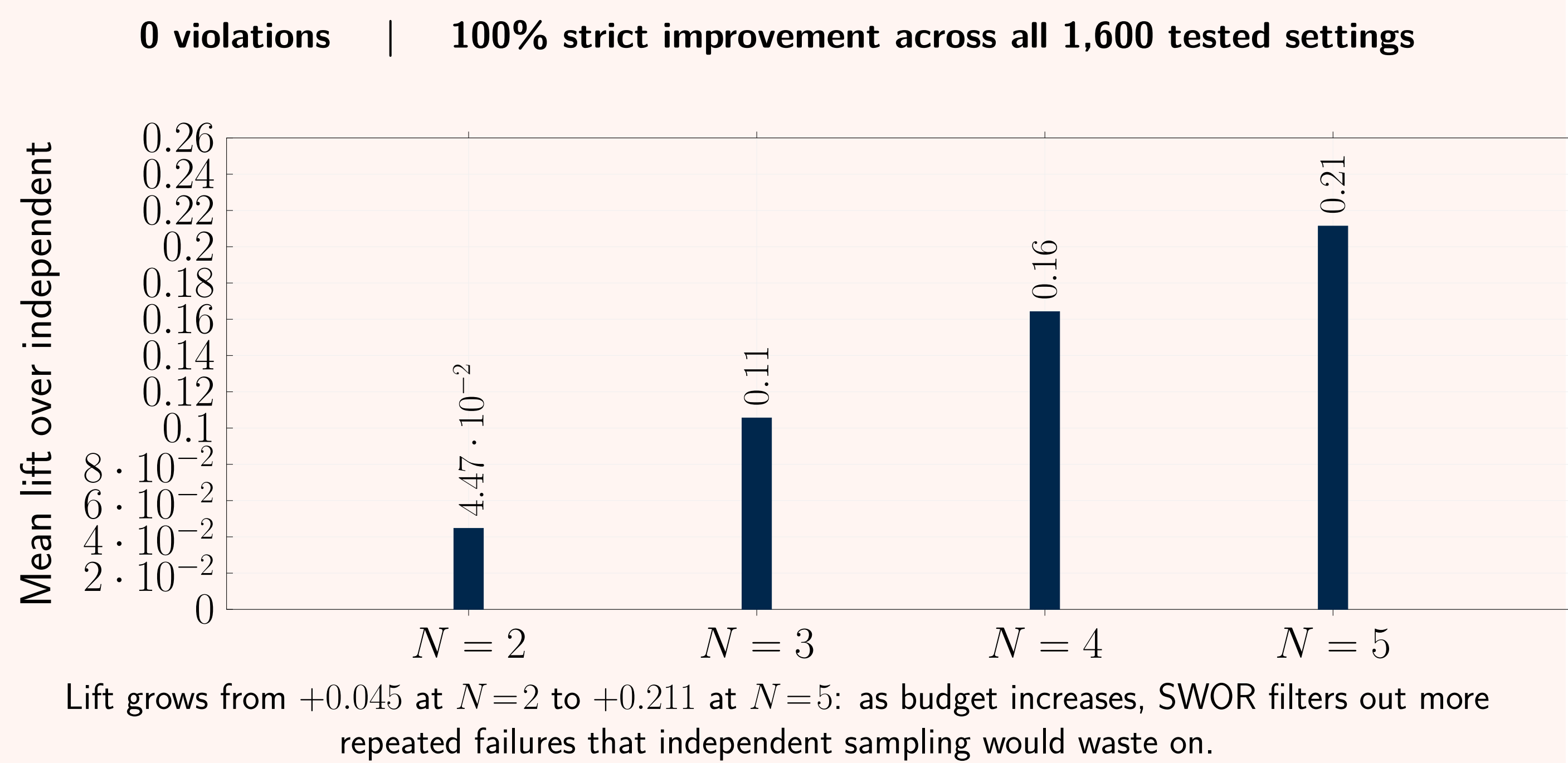


Higher rival concentration (lower α) consistently improves conditional pass@2, confirming the theoretical prediction above.

Pearson $r(\text{HHI, cond. pass@2})$	
$p = 0.05$	0.80
$p = 0.10$	0.89
$p = 0.20$	0.94
$p = 0.35$	0.97

Extreme case ($p = 0.10$)	
Independent pass@2	0.190
Uniform rivals ($\alpha \rightarrow \infty$)	0.201
Single dominant rival	1.000

Result 2: Conditional pass@N Strictly Dominates Independent



What Makes Conditional Search Work?

Our operator-level experiments identify two key distributional properties that determine the value of conditional test-time search:

1. Concentrated wrong-answer mass

When incorrect probability mass is concentrated on a few dominant rivals, eliminating them via SWOR creates large renormalization gains for the correct answer. With diffuse errors (many equally unlikely wrong answers), the gain shrinks.

2. Avoiding overconfidence

Standard CE training pushes models toward high-confidence single answers, collapsing the output diversity that pass@N requires [2]. Chen et al. show that pass@N *decreases* with longer CE training for $N \geq 16$, despite pass@1 increasing. Limiting model confidence rescues pass@N performance.

Implication for training

Test-time-aligned training objectives should re-weight gradients by each sample’s *marginal utility* to the aggregate search outcome, suppressing updates on already-solved problems and reallocating capacity to harder ones. This suggests a promising direction for future search-aware training methods.

Summary of Contributions

1. **Pass@N improves over single sampling**: even on GPT-2 / MATH, generating multiple samples finds solutions that single decoding misses.
2. **Prompt-based conditional sampling shows promise but faces challenges**: the Pink Elephant Effect and context-window noise make “avoid prior answers” conditioning difficult, especially for weaker models [1]. Larger-scale experiments are needed.
3. **The search operator itself is provably beneficial**: under exact SWOR, conditional pass@N strictly dominates independent pass@N across all 1,600 tested settings (4 budgets $\times$ 4 p $\times$ 5 α $\times$ 400 trials).
4. **Concentration of wrong answers is the key condition**: Pearson correlation between rival HHI and conditional pass@2 reaches 0.97; this quantifies when conditional search is most valuable.

Future Directions

- Design training objectives aligned with test-time search strategies and evaluate whether conditional pass@N improves faster than both standard SFT and the pass@N-aware SFT of [2].
- Move beyond prompt-level conditioning: implement SWOR-style filtering at the **logit level** or via **rejection sampling** to bypass instruction-following limitations.
- Scale experiments to stronger models and larger test sets to confirm the trends observed in Experiment 1.

References

- [1] L. Castricato, N. Lile, S. Anand, H. Schoelkopf, S. Verma, and S. Biderman. Suppressing pink elephants with direct principle feedback. *arXiv preprint arXiv:2402.07896*, 2024.
- [2] F. Chen, A. Raventós, N. Cheng, S. Ganguli, and S. Druckmann. Rethinking fine-tuning when scaling test-time compute: Limiting confidence improves mathematical reasoning. *arXiv preprint arXiv:2502.07154*, 2025.